

CHESAPEAKE, VA, USA

WMO: 724006

Lat: 36.666N

Lon: 76.321W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 03701

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.3	-18.0	0.8	-0.6	-14.9	1.0	-0.8	11.0	12.1	9.3	11.2	1.6	330	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	35.0	25.7	33.5	25.2	32.4	24.8	27.5	32.0	26.8	31.2	26.1	30.3	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	29.6	25.9	21.3	29.4	24.8	19.9	28.5	87.0	31.9	83.4	30.9	81.0	30.0	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.5	DB	-11.2	36.8	3.8	1.4	-13.9	37.9	-16.2	38.7	-18.3	39.5	-21.0	40.6	(4)
			WB	-11.9	28.6	3.6	1.3	-14.4	29.5	-16.5	30.3	-18.5	31.0	-21.1	32.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.1	6.2	9.7	15.5	20.2	24.6	26.4	25.6	22.7	17.1	11.2	7.7	(6)
(7)		DBStd	8.82	6.12	5.33	5.50	5.10	4.30	3.20	2.45	2.52	3.06	4.69	5.09	5.46	(7)
(8)		HDD10.0	553	176	131	78	10	0	0	0	0	3	44	111		(8)
(9)		HDD18.3	1810	411	341	273	114	33	1	0	0	4	81	221	331	(9)
(10)		CDD10.0	2762	23	25	68	176	316	438	507	484	382	223	81	41	(10)
(11)		CDD18.3	977	1	1	4	29	91	189	248	226	137	42	7	3	(11)
(12)		CDH23.3	9212	1	6	47	299	821	1849	2562	2152	1102	321	41	11	(12)
(13)	CDH26.7	3595	0	0	7	85	286	786	1120	889	343	75	3	1	(13)	
(14)	Wind	WSAvg	2.4	2.7	3.0	3.1	3.1	2.5	2.1	1.9	1.6	2.1	2.0	2.2	2.3	(14)
(15)	Precipitation	PrecAvg	1281	93	76	102	88	98	115	141	137	137	99	90	90	(15)
(16)		PrecMax	1621	162	199	240	167	140	295	239	327	381	215	261	199	(16)
(17)		PrecMin	800	27	25	20	25	36	39	47	41	15	1	5	36	(17)
(18)		PrecStd	222	39	42	49	38	30	58	53	70	90	52	63	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	24.2	27.5	32.4	34.0	37.1	37.1	36.3	33.1	31.3	27.2	24.9	(19)
(20)			MCWB	18.1	18.1	17.2	21.6	24.7	26.2	26.5	26.7	24.6	23.3	20.7	18.6	(20)
(21)		2%	DB	21.0	21.7	24.1	28.8	32.1	34.2	34.7	34.0	32.0	28.6	24.0	22.1	(21)
(22)			MCWB	17.1	15.7	16.6	19.5	23.2	25.1	26.0	25.6	24.4	22.6	18.6	18.5	(22)
(23)		5%	DB	17.8	18.8	22.3	27.2	29.9	32.7	33.0	32.7	30.1	27.1	22.3	19.1	(23)
(24)			MCWB	14.6	14.8	15.7	18.6	22.1	24.4	25.4	25.4	23.4	21.3	18.0	16.8	(24)
(25)			10%	DB	15.3	16.3	19.8	24.1	27.8	31.2	32.2	31.3	28.6	24.8	20.0	17.4
(26)	MCWB	12.1		11.8	15.3	17.6	21.2	23.7	25.2	24.9	22.9	20.6	16.5	14.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.6	20.8	23.4	26.9	28.5	28.7	28.5	27.1	26.1	22.3	21.0	(27)
(28)			MCD B	21.7	22.5	23.1	28.9	32.3	33.5	34.0	33.5	30.0	27.9	24.3	23.7	(28)
(29)		2%	WB	17.6	17.7	18.8	21.7	24.6	26.6	27.5	27.4	25.9	24.1	20.3	19.0	(29)
(30)			MCD B	19.2	19.9	22.3	26.2	29.5	31.7	32.3	31.8	28.6	26.7	23.1	21.0	(30)
(31)		5%	WB	15.4	15.6	17.2	20.5	23.3	25.6	26.8	26.7	25.1	22.9	18.7	17.4	(31)
(32)			MCD B	17.8	18.4	20.7	24.4	27.5	30.8	31.5	30.8	28.4	25.2	21.3	18.7	(32)
(33)			10%	WB	12.7	12.3	15.5	19.1	22.3	24.8	26.0	25.8	24.3	21.6	17.2	15.2
(34)	MCD B	14.6		15.4	19.1	23.1	26.5	29.6	30.5	29.7	27.5	24.0	19.2	17.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.9	11.5	12.0	12.6	11.4	10.9	10.4	10.3	10.4	11.8	12.2	11.3	(35)
(36)			MCD BR	14.1	14.9	15.5	15.0	13.3	12.6	11.6	11.7	12.2	13.1	13.5	11.9	(36)
(37)			MCWBR	10.9	10.1	9.3	7.1	6.1	5.1	4.9	5.2	6.0	7.4	8.6	9.3	(37)
(38)		5% WB	MCD BR	13.0	14.2	12.4	12.3	11.4	11.3	10.4	10.2	9.4	10.3	10.9	11.3	(38)
(39)			MCWBR	11.4	11.7	9.4	7.4	6.2	5.0	5.0	5.1	5.1	6.6	8.4	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.322	0.349	0.390	0.434	0.485	0.510	0.502	0.406	0.358	0.325	0.314	(40)	
(41)		taud	2.532	2.490	2.444	2.342	2.248	2.160	2.104	2.111	2.376	2.500	2.538	2.549	(41)	
(42)		Ebn at Noon	889	917	918	892	853	807	784	780	847	864	861	859	(42)	
(43)		Edh at Noon	79	92	106	124	138	151	158	154	112	91	78	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.38	3.13	4.17	5.32	5.96	6.37	6.06	5.46	4.46	3.57	2.68	2.12	(44)	
(45)		RadStd	0.19	0.34	0.32	0.39	0.49	0.39	0.30	0.32	0.35	0.41	0.21	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-1.15	-1.35	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+168	+108

Nomenclature: See separate page